UPDATED IMPLEMENTATION FOR ZURICH MODEL BEHAVIOUR REGULATION MODULE

1. Trait-to-Prompt Mappings

Each Big Five trait has **three** possible triggers (+1, –1, 0). We update our previous prompts to make them **concise, precise, and ground them to Zurich Model** ideas of security, arousal, and power, as well as coping/affiliation.

1. **Openness**
   - **+1**: “Invite exploration and novelty.”
   - **–1**: “Focus on familiar topics; reduce novelty.”
2. **Conscientiousness**
   - **+1**: “Provide organized, structured guidance.”
   - **–1**: “Keep the demeanour flexible, relaxed, and spontaneous.”
3. **Extraversion**
   - **+1**: “Maintain an energetic, sociable tone.”
   - **–1**: “Adopt a calm, low-key style with reflective space.”
4. **Agreeableness**
   - **+1**: “Show warmth, empathy, and collaboration.”
   - **–1**: “Use a neutral, matter-of-fact stance; limit personal bonding.”
5. **Neuroticism**
   - **+1**: “Reassure stability and confidence.” *(User is calm/secure)*
   - **–1**: “Offer extra comfort; acknowledge anxieties.” *(User is more anxious)*

Note: These phrases are each grounded in **Zurich Model** insights:

- *“Basic Needs Drive Behavior” (e.g., security ↔ neuroticism, arousal ↔ openness/extraversion, affiliation ↔ agreeableness).*
- *“Graded Responses Instead of Binary Choices” (We keep it short but parse each trait as +1 or –1).*
- *“Harmonize Conflicting Prompts” (We simply list them, expecting future “fixing” if needed).*

2. All Non-Zero Combinations

We now list **all 32 combinations (2^5)** of (+1 or –1) for each trait **O, C, E, A, N**. When a detected trait is 0 (the detection prompts takes care of this) due to lack of information , nothing is added for that dimension. The **final prompt** is a concatenation of whichever trait prompts apply to the **Emotional support assistant baseline behaviour prompt**.

For clarity:

- **O** = Openness (+1 or –1)
- **C** = Conscientiousness (+1 or –1)
- **E** = Extraversion (+1 or –1)
- **A** = Agreeableness (+1 or –1)
- **N** = Neuroticism (+1 or –1)

We show them in a systematic order to ensure that each combination is coherent (**Each negative is in bold** for easy verification):

1) O=+1, C=+1, E=+1, A=+1, N=+1

Final Prompt:

> Invite exploration and novelty;
> Provide organized, structured guidance;
> Maintain an energetic, sociable tone;
> Show warmth, empathy, and collaboration
> Reassure stability and confidence.

2) O=+1, C=+1, E=+1, A=+1, N=–1

> Invite exploration and novelty
> Provide organized, structured guidance
> Maintain an energetic, sociable tone
> Show warmth, empathy, and collaboration
> **Offer extra comfort; acknowledge anxieties.**

3) O=+1, C=+1, E=+1, A=–1, N=+1

> Invite exploration and novelty
> Provide organized, structured guidance
> Maintain an energetic, sociable tone
> **Use a neutral, matter-of-fact stance; limit personal bonding**
> Reassure stability and confidence.

4) O=+1, C=+1, E=+1, A=–1, N=–1

> Invite exploration and novelty
> Provide organized, structured guidance
> Maintain an energetic, sociable tone
> **Use a neutral, matter-of-fact stance; limit personal bonding**
> **Offer extra comfort; acknowledge anxieties.**

5) O=+1, C=+1, E=–1, A=+1, N=+1

> Invite exploration and novelty
> Provide organized, structured guidance
> **Adopt a calm, low-key style with reflective space**
> Show warmth, empathy, and collaboration
> Reassure stability and confidence.

6) O=+1, C=+1, E=–1, A=+1, N=–1

> Invite exploration and novelty;
> Provide organized, structured guidance;
> **Adopt a calm, low-key style with reflective space**
> Show warmth, empathy, and collaboration;
> **Offer extra comfort; acknowledge anxieties.**

7) O=+1, C=+1, E=–1, A=–1, N=+1

> Invite exploration and novelty
> Provide organized, structured guidance
> **Adopt a calm, low-key style with reflective space**
> **Use a neutral, matter-of-fact stance; limit personal bonding**
> Reassure stability and confidence.

8) O=+1, C=+1, E=–1, A=–1, N=–1

> Invite exploration and novelty
> Provide organized, structured guidance
> **Adopt a calm, low-key style with reflective space**
> **Use a neutral, matter-of-fact stance**
> **limit personal bonding; Offer extra comfort; acknowledge anxieties.**

9) O=+1, C=–1, E=+1, A=+1, N=+1

> Invite exploration and novelty
> **Keep the demeanour flexible, relaxed, and spontaneous**
> Maintain an energetic, sociable tone
> Show warmth, empathy, and collaboration
> Reassure stability and confidence.

10) O=+1, C=–1, E=+1, A=+1, N=–1

> Invite exploration and novelty;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> Maintain an energetic, sociable tone;
> Show warmth, empathy, and collaboration;
> **Offer extra comfort; acknowledge anxieties.**”

11) O=+1, C=–1, E=+1, A=–1, N=+1

> Invite exploration and novelty;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> Maintain an energetic, sociable tone;
> **Use a neutral, matter-of-fact stance; limit personal bonding**;
> Reassure stability and confidence.”

12) O=+1, C=–1, E=+1, A=–1, N=–1

> Invite exploration and novelty;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> Maintain an energetic, sociable tone;
> **Use a neutral, matter-of-fact stance; limit personal bonding**;
> **Offer extra comfort; acknowledge anxieties.**”

13) O=+1, C=–1, E=–1, A=+1, N=+1

> Invite exploration and novelty;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> **Adopt a calm, low-key style with reflective space**;
> Show warmth, empathy, and collaboration;
> Reassure stability and confidence.”

14) O=+1, C=–1, E=–1, A=+1, N=–1

> Invite exploration and novelty;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> **Adopt a calm, low-key style with reflective space**;
> Show warmth, empathy, and collaboration;
> **Offer extra comfort; acknowledge anxieties.**”

15) O=+1, C=–1, E=–1, A=–1, N=+1

> Invite exploration and novelty;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> **Adopt a calm, low-key style with reflective space**;
> **Use a neutral, matter-of-fact stance; limit personal bonding**;
> Reassure stability and confidence.”

16) O=+1, C=–1, E=–1, A=–1, N=–1

> Invite exploration and novelty;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> **Adopt a calm, low-key style with reflective space**;
> **Use a neutral, matter-of-fact stance; limit personal bonding**;
> **Offer extra comfort; acknowledge anxieties.**

17) O=–1, C=+1, E=+1, A=+1, N=+1

> **Focus on familiar topics; reduce novelty**;
> Provide organized, structured guidance;
> Maintain an energetic, sociable tone;
> Show warmth, empathy, and collaboration;
> Reassure stability and confidence.

18) O=–1, C=+1, E=+1, A=+1, N=–1

> **Focus on familiar topics; reduce novelty**;
> Provide organized, structured guidance;
> Maintain an energetic, sociable tone;
> Show warmth, empathy, and collaboration;
> **Offer extra comfort; acknowledge anxieties.**”

19) O=–1, C=+1, E=+1, A=–1, N=+1

> **Focus on familiar topics; reduce novelty**;
> Provide organized, structured guidance;
> Maintain an energetic, sociable tone;
> **Use a neutral, matter-of-fact stance; limit personal bonding**;
> Reassure stability and confidence.”

20) O=–1, C=+1, E=+1, A=–1, N=–1

> **Focus on familiar topics; reduce novelty**;
> Provide organized, structured guidance;
> Maintain an energetic, sociable tone;
> **Use a neutral, matter-of-fact stance; limit personal bonding**;
> **Offer extra comfort; acknowledge anxieties.**”

21) O=–1, C=+1, E=–1, A=+1, N=+1

> **Focus on familiar topics; reduce novelty**;
> Provide organized, structured guidance;
> **Adopt a calm, low-key style with reflective space**;
> Show warmth, empathy, and collaboration;
> Reassure stability and confidence.”

22) O=–1, C=+1, E=–1, A=+1, N=–1

> **Focus on familiar topics; reduce novelty**;
> Provide organized, structured guidance;
> **Adopt a calm, low-key style with reflective space**;
> Show warmth, empathy, and collaboration;
> **Offer extra comfort; acknowledge anxieties.**”

23) O=–1, C=+1, E=–1, A=–1, N=+1

> **Focus on familiar topics; reduce novelty**;
> Provide organized, structured guidance;
> **Adopt a calm, low-key style with reflective space;**
> **Use a neutral, matter-of-fact stance; limit personal bonding**;
> Reassure stability and confidence.”

24) O=–1, C=+1, E=–1, A=–1, N=–1

> **Focus on familiar topics; reduce novelty**;
> Provide organized, structured guidance;
> **Adopt a calm, low-key style with reflective space;**
> **Use a neutral, matter-of-fact stance; limit personal bonding**;
> **Offer extra comfort; acknowledge anxieties.**

25) O=–1, C=–1, E=+1, A=+1, N=+1

> **Focus on familiar topics; reduce novelty**;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> Maintain an energetic, sociable tone;
> Show warmth, empathy, and collaboration;
> Reassure stability and confidence.”

26) O=–1, C=–1, E=+1, A=+1, N=–1

> **Focus on familiar topics; reduce novelty**;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> Maintain an energetic, sociable tone;
> Show warmth, empathy, and collaboration;
> **Offer extra comfort; acknowledge anxieties.**”

27) O=–1, C=–1, E=+1, A=–1, N=+1

> **Focus on familiar topics; reduce novelty**;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> Maintain an energetic, sociable tone;
> **Use a neutral, matter-of-fact stance; limit personal bonding**;
> Reassure stability and confidence.”

28) O=–1, C=–1, E=+1, A=–1, N=–1

> **Focus on familiar topics; reduce novelty**;
> **Keep the demeanour flexible, relaxed, and spontaneous**;
> Maintain an energetic, sociable tone;
> **Use a neutral, matter-of-fact stance; limit personal bonding; **
> **Offer extra comfort; acknowledge anxieties.**

29) O=–1, C=–1, E=–1, A=+1, N=+1

> **Focus on familiar topics; reduce novelty**;
> **Keep the demeanour flexible, relaxed, and spontaneous;**
> **Adopt a calm, low-key style with reflective space**;
> Show warmth, empathy, and collaboration;
> Reassure stability and confidence."

30) O=–1, C=–1, E=–1, A=+1, N=–1

> **Focus on familiar topics; reduce novelty**;
> **Keep the demeanour flexible, relaxed, and spontaneous;
> Adopt a calm, low-key style with reflective space; **
> Show warmth, empathy, and collaboration;
> **Offer extra comfort; acknowledge anxieties.**

31) O=–1, C=–1, E=–1, A=–1, N=+1

> **Focus on familiar topics; reduce novelty**;
> **Keep the demeanour flexible, relaxed, and spontaneous;
> Adopt a calm, low-key style with reflective space;
> Use a neutral, matter-of-fact stance; limit personal bonding; **
> Reassure stability and confidence."

32) O=–1, C=–1, E=–1, A=–1, N=–1

> **Focus on familiar topics; reduce novelty**;
> **Keep the demeanour flexible, relaxed, and spontaneous;**
> **Adopt a calm, low-key style with reflective space;**
> **Use a neutral, matter-of-fact stance; limit personal bonding;**
> **Offer extra comfort; acknowledge anxieties.**

3. Using These Prompts

- **Implementation**: When an OCEAN detection is updated (for each trait ±1), the system **concatenates** the relevant short behaviour regulation prompts. If a trait is 0, nothing is added for that dimension.
- **Zurich Model Alignment**:
  - *"Personality is Dynamic"*: Each detection shift can alter the prompts in real time.
  - *"Interdependent Coping Strategies"*: For example, if someone is anxious (N=–1) and unstructured (C=–1), the system responds with more supportive, flexible language.

4. Benefits of This Simple Mapping Approach

1. **Progressively More Accurate Personality Detection**
   - Because the detection module updates OCEAN scores after every new user message (incorporating the entire chat history), personality estimates become **incrementally more accurate** as the conversation unfolds.
2. **Transparent and Traceable Logic**
   - By **directly mapping** trait values (± 1) to short behavior prompts (rather than formulas), we reduce "black-box" complexity.
   - This makes it **easier to debug**, **explain to stakeholders**, and **modify** if any prompt feels off or contradictory.
3. **Easy for Iterative Development**
   - Developers can **quickly adjust** or **fine-tune** prompts without overhauling a complex formula layer.
   - If new insights or test data show a need to refine how a particular trait influences the Assistant's tone, it's just **one quick change** in the prompt mapping.
4. **Flexible for Other Real-Time Signals**
   - The same approach can accommodate **other frameworks**, such as real-time emotion detection.
   - You simply swap "if OCEAN = ± 1" for "if user sentiment is positive/negative, " or any other detection system, and **attach suitable prompts**.

CODE IMPLEMENTATION

Note: This is only for inspiration, the implementation will need to take into account how the original implemented classes work.

Below is a **straightforward** way (**Please check the PROMISE implementation first for compatibility**) to store these mappings in Java using **enums** and **maps**. This setup makes it easy to:

1. **Look up** each trait's prompt for `+1` or `-1`.
2. **Ignore** the trait if its value is `0` (no prompt).

You can then **concatenate** the resulting prompts into the final behavior string.

1. Define an Enum for Each OCEAN Trait

```
public enum OceanTrait {
    OPENNESS,
    CONSCIENTIOUSNESS,
    EXTRAVERSION,
    AGREEABLENESS,
    NEUROTICISM
}
```

2. Store Prompts in a Nested Map

Here, the **outer map** keys are your `OceanTrait` values, and the **inner map** keys are `Integer` trait scores (+1 or –1). The **value** in the inner map is the **prompt** string.

```java
import java.util.HashMap;
import java.util.Map;

public class OceanPrompts {

    private static final Map<OceanTrait, Map<Integer, String>> PROMPT_MAP = new HashMap<>();

    static {
        // For Openness
        Map<Integer, String> opennessMap = new HashMap<>();
        opennessMap.put(+1, "Invite exploration and novelty.");
        opennessMap.put(-1, "Focus on familiar topics; reduce novelty.");
        PROMPT_MAP.put(OceanTrait.OPENNESS, opennessMap);

        // For Conscientiousness
        Map<Integer, String> conscMap = new HashMap<>();
        conscMap.put(+1, "Provide organized, structured guidance.");
        conscMap.put(-1, "Keep the demeanour flexible, relaxed, and spontaneous.");
        PROMPT_MAP.put(OceanTrait.CONSCIENTIOUSNESS, conscMap);

        // For Extraversion
        Map<Integer, String> extraMap = new HashMap<>();
        extraMap.put(+1, "Maintain an energetic, sociable tone.");
        extraMap.put(-1, "Adopt a calm, low-key style with reflective space.");
        PROMPT_MAP.put(OceanTrait.EXTRAVERSION, extraMap);

        // For Agreeableness
        Map<Integer, String> agreeMap = new HashMap<>();
        agreeMap.put(+1, "Show warmth, empathy, and collaboration.");
        agreeMap.put(-1, "Use a neutral, matter-of-fact stance; limit personal bonding.");
        PROMPT_MAP.put(OceanTrait.AGREEABLENESS, agreeMap);

        // For Neuroticism
        Map<Integer, String> neuroMap = new HashMap<>();
        neuroMap.put(+1, "Reassure stability and confidence.");
        neuroMap.put(-1, "Offer extra comfort; acknowledge anxieties.");
        PROMPT_MAP.put(OceanTrait.NEUROTICISM, neuroMap);
    }

    // Public accessor
    public static String getPrompt(OceanTrait trait, int value) {
        // Return the matching prompt, or null if no match
        Map<Integer, String> traitMap = PROMPT_MAP.get(trait);
```

```
        if (traitMap == null) {
            return null;
        }
        return traitMap.get(value);
    }
}
```

3. Generating the Final Prompt

When the detection module **detects** an updated OCEAN tuple, for example (o, c, e, a, n), you can do:

```java
public class BehaviorRegulator {

    // Example method: build final prompt for an OCEAN tuple
    public static String buildPrompt(
        int opennessValue,
        int conscientiousnessValue,
        int extraversionValue,
        int agreeablenessValue,
        int neuroticismValue
    ) {
        StringBuilder sb = new StringBuilder();

        // For each trait, if the integer != 0, look up the prompt
        if (opennessValue != 0) {
            String prompt = OceanPrompts.getPrompt(OceanTrait.OPENNESS, opennessValue);
            if (prompt != null) sb.append(prompt).append(" ");
        }
        if (conscientiousnessValue != 0) {
            String prompt = OceanPrompts.getPrompt(OceanTrait.CONSCIENTIOUSNESS, conscientiousne
            if (prompt != null) sb.append(prompt).append(" ");
        }
        if (extraversionValue != 0) {
            String prompt = OceanPrompts.getPrompt(OceanTrait.EXTRAVERSION, extraversionValue);
            if (prompt != null) sb.append(prompt).append(" ");
        }
        if (agreeablenessValue != 0) {
            String prompt = OceanPrompts.getPrompt(OceanTrait.AGREEABLENESS, agreeablenessValue
            if (prompt != null) sb.append(prompt).append(" ");
        }
        if (neuroticismValue != 0) {
            String prompt = OceanPrompts.getPrompt(OceanTrait.NEUROTICISM, neuroticismValue);
            if (prompt != null) sb.append(prompt).append(" ");
        }

        return sb.toString().trim();
    }

    // Example usage
    public static void main(String[] args) {
        // Suppose we detected a user with O=-1, C=+1, E=+1, A=-1, N=+1
        String finalPrompt = buildPrompt(-1, +1, +1, -1, +1);
        System.out.println(finalPrompt);
        // Output (example):
```

```
        // Focus on familiar topics; reduce novelty. Provide organized, structured guidance.
        // Maintain an energetic, sociable tone. Use a neutral, matter-of-fact stance; limit per
        // Reassure stability and confidence.
    }
}
```

1. **Check if a trait value is `0`**. If so, **skip** adding any text.
2. **Concatenate** non-null prompts into a final string.
3. Return or display it to the user.

Why This Setup

- **Transparency**: Each trait's text is explicitly in the `PROMPT_MAP`.
- **Maintainability**: If you want to **edit** a single prompt (e.g., changing wording for *Openness +1*), you do it in **one place**.
- **Modularity**: The dictionary approach can easily be extended or replaced (e.g., new frameworks for emotion detection) without refactoring your entire codebase.
- **Iterative**: You can **tweak** or add partial logic (like conflict resolution) later in a well-structured manner.